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Human, technological and organizational aspects influencing the production scheduling process

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Abstract

This study of scheduling work in practice addresses how the production-scheduling processes in four companies are influenced by human, technological, and organizational aspects. A conclusion is that the outcome of the scheduling process is influenced by the scheduler adding human capabilities that cannot be automated, by technical constraints in the scheduled production system and by the available scheduling software tools. Furthermore, the outcome is influenced not only by how the scheduling process is formally organized, but also by the scheduler's informal authority and the role taken to interconnect activities between different organizational groups. The findings from the study support a number of previous studies done on scheduling in practice whilst giving new insights into their interpretation.
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1. Introduction

Production scheduling is an activity concerning fulfilling deliveries in which many demands and restrictions must be handled and negotiated. The current trend towards reductions in lead times means that the scheduling tasks must also be performed in continuously reducing time periods. At the same time, reductions of inventory, and the removal of material buffers in situations where customers demand improved service, make the fulfilment of customer demands more challenging. These conditions make production scheduling increasingly difficult to handle for many enterprises. To succeed in coping with these difficulties, improvements in production scheduling activities is critical (Vollman et al., 1997).

The dominant research and development emphasis in scheduling has been aimed mainly at solving technical, logical or mathematical problems. The research reported here has another perspective with a focus on what scheduling in practice actually encompasses. It contributes to a research stream that has been describing and developing knowledge about scheduling from the perspective of the human scheduler (Sanderson, 1989; Wiers and Van Der Schaaf, 1997; Crawford et al., 1999; McKay and Wiers, 1999; Berglund and Karltun, 2000). In addition scheduling has been found to be influenced by aspects such as roles, trust, respect and interpersonal interaction (McKay et al., 1992). Further studies have highlighted the complexity of

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planning and scheduling, and that it is comprised of and influenced by technical, organizational and human aspects (MacCarthy and Wilson, 2001). Furthermore, MacCarthy et al. (2001) identified the skill and performance of production schedulers as making a significant contribution to the achievement of high productivity and flexibility.

The results presented in this paper are part of a study of production planning and scheduling practice undertaken in four Swedish woodworking companies: a sawmill, a parquet manufacturer, a furniture manufacturer, and a house manufacturer. Earlier work drawn from the analysis of these cases has addressed expectations on schedulers (Berglund and Karltun, 2001), how schedulers manage goal conflicts (Karltun and Berglund, 2002a) and the ways in which they influence other employees' working conditions (Karltun and Berglund, 2002b). This paper takes a new perspective, investigating how the production scheduling processes are influenced by human, technological and organizational aspects. In addition to generating findings in this area, it draws some conclusions on the extent to which this perspective can generate an improved understanding of the schedulers' work.

The paper first provides a description of the humans, technology and organization (HTO-) concept, which serves both as a meta-concept and a model for analysis that takes into account humans, technology and organization to analyse and improve work systems (Westlander, 1999a). It describes how the concept is applied to the analysis of scheduling, categorising aspects of scheduling into the three components, and highlighting key issues from the analysis. Finally, the identified issues, the value of HTO as a concept for understanding the scheduling process, and the influence of HTO components on the investigated production scheduling processes are presented and discussed.

The distinction between planning and scheduling is somewhat unclear. One definition is that scheduling is associated with decisions made when the production process is running, while planning is associated with decisions made before the production process starts (Nakamura and Salvendy, 1994). However, as the precise interaction between production planning and scheduling is not easily defined it may be justified to assume that planning and scheduling in fact represent a continuum of activities across space and time (Crawford et al., 1999). Dispatching may also be included in this continuum of activities. It deals with deciding which operation to perform next at an individual workplace, but there is overlap between scheduling and dispatching (McKay and Wiers, 2003). Overall, the work activities studied are the activities dealing with capacity, scheduling of production, as well as dispatching individual orders. In these case studies, as production scheduling constituted the largest proportion of these activities, the term "scheduling" is used throughout the paper.

2. Humans, technology and organization

As the scheduling process is closely linked to the production department, and also to the sales department (Karltun and Berglund, 2002a), we propose the use of a systems perspective (von Bertalanffy, 1968; Checkland, 1981) to reach a deeper understanding of the scheduling process. There are various aims associated with using a systems perspective, such as improving both organizational performance and working conditions, as in sociotechnical systems theory (Pasmore, 1988), or as means for organizational development (Porras and Robertson, 1992).

The HTO-concept was developed during the 1980s within the nuclear power industry. The aim was to improve overall safety. When developing safety, strong focus was first put on technical improvements, which resulted in a reduced number of near-accident incidents caused by technical failures. As a result, the incidents caused by humans became more "visible" as they represented a larger proportion of the reported incidents. After a period focusing on the field of potential "human error", safety was further improved. At that time it became apparent that it was also necessary to consider organizational issues. This led to the realization that all three HTO components in the system need to be addressed to improve safety (Eklund, 2003).

The system view of the HTO-concept was considered successful for improving safety and developing a more thorough safety culture and has thereafter spread to other domains (Rollenhagen, 1997, 2003). The HTO-concept has since reached wider use in Sweden as a meta-concept that may be applied to analyse and further develop understanding of highly complex work activities. The main point is that successful development is only achieved if all three components are considered together. See Fig. 1 for an overview of the components. The concept is also valuable in research as it ensures that a broad range of factors are considered. Where a specific focus is chosen for research, for instance

focusing on technology, human and organizational aspects may be analysed and treated as influencing factors (Westlander, 1999a).

The HTO-concept distinguishes between the individuals and the organization. Therefore, it is suitable for the study of individual scheduler's work. In the HTO-perspective, the "H" stands for the importance of aspects that are strictly individual and at the same time considered important to perform a task or a change. Such aspects may, for example, include individual skill, knowledge, experiences or established relations with other people. The H-component focuses on the individual's contribution to the business process. It may be described at any of the following four levels (Daniellou, 2001).

(1) the biological level, regarding the human as a physiological system,
(2) the cognitive level, where the human is considered as an information-processing system, including thought processes, representations, and decision-making,
(3) the psychological level, where the human has a unique history, leading to a specific subjective processing of the situations he/she experiences, and
(4) the social level, emphasizing that every single individual is a member of several social groups with different cultures, which will partly determine his/her values and habits.

In the case of investigating scheduling work, the human scheduler was considered at the cognitive, the psychological and the social level.

The "T" in HTO, stands for the technical system. This can be divided into two parts. The first is the *primary technical system* that includes the production equipment that is devoted to maintaining the capability of the company. This technical system can be described in different ways in relation to technical limitations, problems (both recurrent and stochastic), availability, and reliability. With regard to complex technical systems, the system may be considered as networks of relatively linked autonomous parts, which are to a large extent separately controlled and managed. A *secondary technical system* is a system that assists the administration and procedures of the company but is not directly associated with the value-adding activities of the business (Wäfler, 2001). In this study, the secondary technical system consists of the information system, hardware and software that schedulers use as decision support tools.

Regarding the HT-interaction, some criteria for the design of the scheduling software system are the level of support, transparency, autonomy and information presentation (Wiers, 2001), also whether it is integrated with the task, tolerant of errors, provides relevant feedback, and provides sufficient decision latitude.

The third component, the "O" in HTO represents the H-component in an aggregated sense. It is not considering single individuals but humans in a statistical interchangeable way (Westlander, 1999a). The O-component comprises how the work is organized and structured, both formally and informally (Westlander, 1999b). Examples of the formal organization are job definitions, responsibilities and powers, hierarchical positions, policies, business goals and strategies. The O-component also includes rules, procedures and cultural factors as well as relations between system components and subsystems, which all have both a formal and informal side.

The interaction between the scheduler and the organization is naturally very extensive and can be described at different levels and from different perspectives. Drawing from role theory, there are role definitions and expectations made at different levels of the organization, including managers, colleagues and other workers. The roles become more complex when they require the individual to be simultaneously involved in two or more subsystems, since each is likely to have its own priorities and to some degree its own subculture (Katz and Kahn, 1978). In the case concerning the schedulers, there is reason to believe that the demands and expectations of them may contribute to the complexity of scheduling practice and impact on the schedulers' performance. Jackson et al. (2004) described the complexity of schedulers' roles and divided these into three main roles: the interpersonal role, the informational role and the decisional role. In addition, there are physical aspects of the O-component in relation to where people are located and how premises are designed (Weisbord, 2004).

In a systems perspective, these components will interact in different ways. The authors' assumption is that many of the interesting characteristics of the schedulers' work are to be found in the interaction between the components. As this paper addresses the work of the human scheduler, it analyses the interactions between the H- and T- and

the H- and O- but ignores those between the T- and O-components.

3. Method

Scheduling work was studied in four companies. These companies were selected according to two criteria; that the companies had different types of production characteristics and that each company had at least one employee regarded as a full time scheduler. The first criterion was set to establish whether the same phenomena in the investigated scheduling work existed at some sites but not at others (Stuart et al., 2002).

The methodological approach chosen for the overall project was ergonomic work analysis (Guérin et al., 1997). It is based on the analysis of work activities carried out in real work situations. It sets out to learn about the employees' knowledge and competence, how information is collected and treated and how decisions are made. The method first establishes a detailed description of the mission, that is why the enterprise agreed to participate in the research project, the scheduling task, and the work context comprising technology, work process, roles and activities. Data collection was carried out in three stages, by acquiring knowledge of scheduling as described initially by the management, as further described in the researchers' interviews with the staff, and through observation of the scheduling activities.

The interviews covered the production scheduling process, the interviewees' involvement and the interactions that took place between different people in different roles. The interviewees chosen were those who the schedulers identified as crucial for their work. Documents, software support, characteristics of the technical production system as well as the type of demand for each product category were studied and described. This data collection provided the authors' with an initial overview and understanding of scheduling work prior to more detailed observations of scheduling activities in practice being conducted. Furthermore, it provided the authors with descriptions of what the company believed scheduling work consisted of within the company. This was crucial in developing an understanding of how the scheduling work was organized and formally structured so providing essential knowledge of the formal side of the O-component.

Each scheduler's work was observed over five entire working days throughout a time period of three weeks for two cases and eight weeks for the other two cases. Observations took place each day of the week (Monday, Tuesday etc.) to allow for regular, weekly variations in work to be observed. During the observations all notable work activities were recorded in terms of descriptions of tasks, timing, the people involved, the use of aids, who took initiatives to activities, the communication patterns employed, and the physical location of the schedulers. The informal organization and structure of scheduling work was registered but also the schedulers' social contacts and work patterns. These observations also provided data on the cognitive and psychological demands placed on the schedulers, which were followed up for validation in interviews. The observations thus gave data on all three H-, T-, and O-components.

After making the observations, a second set of interviews was carried out, in which the interviewees evaluated the state of scheduling practice in their company, their expectations on the scheduler, how their work was affected by the scheduler, and how they themselves affected the work of the scheduler. The interviewees selected for this second part focused on the schedulers and the other staff identified as having close contact with the scheduler during the observations. The participants interviewed after the observations are presented in Table 1.

To assess the value of applying an HTO-perspective the data collected were analysed in relation to the different components and their interactions. The interaction between the components "O" and "T" was omitted as the collected data did not support that analysis. Furthermore, the interaction between the schedulers and their decision support systems was conducted only at a cursory level.

The empirical data were analysed in a number of ways. The data from the initial interviews and documents were reported to provide an overview of each company's business activities and the ways scheduling was supposed to be performed. The data from the second set of interviews were transcribed into tables where the answers for each question were compared between interviewees and companies.

The data from the observations were analysed qualitatively and quantified. For each case observations were elaborated separately with all the work activities undertaken by the scheduler being described

Table 1
Interviews after the observations

Company	No of interviewees	Interviewees' position
Sawmill	15	Scheduler, purchaser, marketing manager, CEO, production manager, planing scheduler, delivery officer, five operators and three forklift truck drivers
Parquet manufacturer	21	Scheduler, four production supervisors, four production group leaders, two forklift truck drivers, long distance truck driver, six customer helpdesk officers, quality coordinator, drying kiln foreman, purchaser
Furniture manufacturer	10	Scheduler, quality manager (in owner family), delivery officer, two foremen, purchaser, production manager, supervisor, two purchasers from the customer
House manufacturer	19	Scheduler, foreman, two supervisors, industrial engineer, production manager, purchaser, material administrator, three delivery officers, two sales support officers, export officer, project coordinator, construction manager, three construction engineers

and categorized. To be able to perform a cross case analysis (Yin, 1994) between the four companies, the main activities for each scheduler were grouped into generalized categories.

Furthermore, all *intervening activities* where the schedulers suddenly changed from the main on-going activities to other activities were listed and categorized into two types. The first type was defined as *interrupting activities* and consisted of activities that were included in the scheduler's normal tasks. The interrupting activities were initiated both by the schedulers themselves and by other people. The second type was *disturbances*, which comprised activities not related to the scheduling task. This could for example be a manager wanting to discuss a social activity for the personnel or someone wanting help with the photocopier.

4. Company descriptions

In this section, the case study companies are briefly described, including the schedulers' formal tasks and responsibilities. The key characteristics of the investigated companies are summarized in Table 2.

4.1. The sawmill

The sawmill belonged to a larger sawmill group. It was organized in two sub-units. One was the

Table 2
Overview of company characteristics

Company	No of employees	Products	Scheduled production system	Scheduling aim
Sawmill	42	Beams, tongue and groove boards, scaffold planks	Sawmill and drying kilns	Optimize the high value outcome of the sawing process regarding quality and volume
Parquet manufacturer	1400	Parquet floor	Production of hard surface of the parquet floor (pressing, formatting and surface treatment lines)	Find best possible fit between production capability and delivery demands
Furniture manufacturer	220	Bookshelves	Entire production	Control and manage the relation between forecast, order situation, schedule and actual deliveries
House manufacturer	370	Wooden houses	Drafting process, block production (pre-assembly at production site), parts delivery and house erection (final assembly at building site)	Capacity utilization and deliveries on time

sawmill with 42 employees, out of which five were white-collar workers, including the sawmill scheduler and the plant manager. The other unit was a secondary woodworking industry performing planing operations and also producing some of its own products such as cable drums and storage shelves. This unit had about 90 employees.

The sawmill produced about 110,000 m^3 sawn wood mainly for the international market. Products manufactured for the building and construction market were beams, tongue and groove boards, and scaffold planks. Forty percent of company production was exported to Japan, with a smaller quantity to Taiwan. The sawmill's primary focus on the Japanese market put new demands on the company, such as producing beams with very fine tolerances and specific shipping conditions.

The planning and scheduling work analysed was that performed at the sawmill. The scheduler's main contribution was to optimize high value output from the sawing process achieving both volume and quality production. Further to this, it included achieving optimal fit between available timber supply and customer demand. In order to reach these goals the scheduler tried to influence supply, customer demand and the production process. The scheduler also took part in controlling the quality grading of produced high-value goods. Furthermore, the scheduler updated the scheduling software system both continuously and on a daily basis. The company aimed to sell most of the sawmill production based on long-term contracts, typically agreed for a 3-month period at a time. The company had its own marketing manager, making the customer contracts. The customer then "called off" the required volume of product within the contract terms.

4.2. The parquet manufacturer

The parquet manufacturer was a major parquet manufacturer on the world market with sales agents in many countries. Its weekly production was more than 100,000 m^2 of parquet. It had about 2000 employees in different plants, most of which were situated in Sweden. This study was carried out in the main plant, which had about 1400 employees of which 1200 employees were in production.

Scheduling involved the co-ordination of this production with that of the other plants. The production scheduling function belonged organizationally to production and consisted of four people, two of whom were involved with daily production and two who worked with the computerized scheduling system and development projects. The scheduler being observed in this study received orders from sales and entered job orders in the scheduling software system. Three main production flows were scheduled within the same premises and parts of production in two other premises. The scheduler's responsibility was comprised of finding the best possible fit between production capability and delivery demands.

4.3. The furniture manufacturer

The furniture manufacturer was a family owned company with 220 employees. It produced 1.3 million bookshelves each year in different models for an international furniture retailer. This production constituted about 97% of company turnover. The company faced strong competition and hard cost pressure, but it had at the time a three-year agreement with the retailer. A large investment in an automated production line had recently been made, which was expected to strongly increase productivity and production volumes. Production set-ups were time consuming; so long production runs were preferred and sought.

The scheduler's contribution to the scheduling process was controlling and managing the balance between order requirements, the schedule and actual deliveries. This included recognizing the current production capability, analysing the order situation as well as the customer forecast. It further included establishing delivery feasibility and dates with the main customer. The scheduler also updated the scheduling software system and was responsible for the functionality of the system in the company.

4.4. The house manufacturer

The house manufacturer produced wooden houses for the Swedish market. It had 370 employees. Two thirds of the employees were white-collar workers and the remaining blue-collar workers. The large proportion of white-collar workers was due to the large amount of design and administrative work associated with building and contracting management. About 90% of all houses sold were delivered, erected and installed by the business at the building site. The main market was Sweden, but the company also delivered to other countries in northern Europe.

From the scheduler's point of view, there were two different processes to manage: single family houses sold to private customers and larger houses or groups of houses sold to individual external building contractors. The company was undergoing a change from delivering a standard house product to more customized houses. The company was, therefore, continuously modifying its organization and production and delivery processes to improve its performance. The scheduler's tasks included making a master and detailed schedule for each house produced, controlling the drafting procedures with respect to necessary scheduling steps, synchronizing three production units, and ordering and planning flows of material to the production units and building sites.

5. Findings

5.1. Human aspects

Apart from the expected cognitive workload imposed by the scheduling task itself, the study showed that scheduling in practice required a great deal of additional cognitive work as it was characterized by abrupt changes from one work task to another, initiated by someone calling, entering the office, discovery of unexpected data in the scheduling software tool, or urgent e-mails. This is illustrated in Table 3, in which one observed main activity for the scheduler at the parquet manufacturer is described including intervening activities. As described earlier some of these consisted of interrupting activities, such as handling inquiries on customer-related issues, production and order status, or technical problems.

Others were defined as disturbances and consisted of the schedulers dealing with computer problems and various inquiries about addresses, other people and materials. During five days of observation, the scheduler at the parquet manufacturer changed from the main activity to intervening activities on 268 occasions. In the case of the scheduler at the furniture manufacturer, there was a total of 98 intervening activities during four days (as he visited the customer during one of the observation days). Although facing approximately half as many intervening activities as the parquet scheduler, the furniture scheduler chose to start an hour and a half before other employees in the morning to be able to work undisturbed with the schedule.

There were numerous examples of several issues being handled at the same time and of a strong need to work in a flexible manner. It was also clear that the human contribution to scheduling practice, at a cognitive level, included a great deal of problem solving. Problem solving was carried out as part of the schedulers' planned activities, but often also as a consequence of something unplanned occurring in production or new information from the market. Problem solving included activities such as handling unexpected events, co-ordinating different production units, and participating in production and product development.

One example of problem solving at the sawmill was when the scheduler in the secondary woodworking industry informed the sawmill scheduler that a salesman had placed an order without checking available stock. As it was not possible to deliver the ordered products, the sawmill scheduler informed the salesman that only a limited quantity could be delivered. Being obviously upset, the scheduler at the same time inquired at a competing sawmill if they could deliver the required quantity and dimensions of boards. The problem was solved this way within a few hours and on the scheduler's own initiative, although he had the possibility to refuse dealing with the order as the salesman had not followed the routines.

It was also found that problem solving had a social dimension. Production personnel, shop floor supervisors, market representatives and the management strongly pointed out in all companies that they expected the production schedulers to solve problems in many areas. The schedulers were expected to have good understanding of the problems, be able to suggest and discuss different solutions, and carry through the solutions chosen. This was expressed by the deputy managing director at the furniture manufacturer as:

> It is very important for the scheduler not to be too afraid, to chase a little, to develop a rather tight schedule and not to spend half a day extra. We are so sensitive to price that we must push on.

The house scheduler also formulated the importance of carrying through solutions as:

> Scheduling is not difficult, but one must dare. Mistakes might result in severe consequences.

The schedulers confirmed in the interviews that they believed that others in the company considered them to be problem solvers, someone to turn to with

Table 3
Example of the scheduler's main activity and intervening activities at the parquet manufacturer

Main activity: Deleting already produced orders from the software system	*Intervening activities:* The scheduler changes from the main activity to deal with the intervening activity for some period of time and returns afterwards to the main activity	
Start of main activity: 13:03	Time	Short description
Stop of main activity: 14:54	13:03	Forklift truck driver 1 enters and states what has actually been produced in the surface oiling line
	13:07	Production leader 1 calls and informs about a disturbance due to rain in the packing line
General description		
The scheduler scans all orders in the scheduling software tool and deletes all production orders that have already been executed. Furthermore, he makes changes in delivery times for some orders where there are disturbances and those without alternative delivery times are put aside for further handling.	13:09	Production leader 2 calls and asks if he should mark that the articles should be oiled or not in the packing line
	13:15	Forklift truck driver 2 enters and seeks forklift truck driver 1 who often visits the scheduler
Other departments are informed about these changes by mails or documents written along with making the changes. In this example the pressing line receives an explanation as well as customer service. In the latter case the changes are due to lack of incoming material and production disturbances.	13:21	Truck driver calls and asks is there is any delivery to another production plant this week
	13:31	Salesman calls with an inquiry
	13:38	Customer service officer 1 calls about certain production order that will be executed although not shown in the software system
	13:51	Economy department wants information about some articles
	14:04	Scheduler colleague enters and asks whether a certain parquet will be produced in a certain week
	14:11	Production foreman enters and asks about the production the following day in the glazing line
The scheduler works in parallel with the scheduling software tool and own developed spreadsheets.	14:15	Customer service officer 2 calls and asks where the scheduler's colleague is (who is not answering on her phone)
At the end of the main activity the scheduler has checked more than 400 orders.	14:16	Scheduler colleague calls and informs that he is now dealing with the software problems that occurred last weekend
	14:21	Customer service officer 3 calls with a product inquiry
	14:33	Scheduler colleague calls out from his next door office and asks about a telephone number

all kinds of questions and problems. This was also noted during the observations in all the companies studied and is illustrated in Table 3 for the parquet scheduler.

Further role expectations on the schedulers expressed in the second set of interviews were that they should be flexible and willing to discuss changes in the schedule when initiated by other employees or the market. This was further emphasized as 88% of the interviewees in all the companies perceived that they were able to influence the schedule.

In relation to the expectation of being flexible, the schedulers were expected to find compromises and negotiate between groups in the company where such groups had incompatible goals and differing logics. One example of this was the schedulers' need to continuously handle and make compromises on different, sometimes incompatible time scales, varying from single hours to two months for fulfilling operations or deliveries. At the sawmill, 21 different time scales were identified that the scheduler had to take into consideration when developing the schedule.

Numerous unanticipated events, with sources both within and outside the company, resulted in a high level of uncertainty. The high degree of uncertainty resulted in the schedulers continuously searching for process feedback and acting to increase their decision latitudes. In many cases, the only way to receive the required information was for the schedulers to visit other offices or production premises.

Whilst visiting other areas they often checked up on problems that occurred earlier, checked that production was running as planned, and visually inspected the levels of finished goods. This patrolling of production was particularly prevalent at the sawmill and the parquet manufacturer. Another example of the scheduler acting to increase decision latitude and control was observed at the parquet manufacturer as the scheduler received an order of a very small quantity of a specific type of parquet. As executing the order would cause a great deal of work in the workshop the scheduler gave the customer a long delivery time and hoped that more similar orders would arrive before scheduling it in the workshop.

Regarding the psychological level in the H-component, all schedulers have had substantial shop floor work experience in their own and/or similar companies. The total work experience from the woodworking industry was 11, 24, 29, and 30 years for each scheduler respectively. The experience they had gained was very useful for performing the scheduling task. For instance the schedulers had thorough knowledge of the machinery and its constraints, and could easily understand the implications of machine failure. This knowledge also gave them good legitimacy within the production units.

Human aspects at a social level included the fact that the schedulers spent a significant amount of time in contact with other people in the organization, such as the production personnel, production supervisors, production engineers, the market department, and forklift truck drivers. In interactions with these people the issues dealt with included production feedback, manning issues, work orders, quality control, development projects, product development, and deliveries.

The schedulers pointed out how important these personal contacts were in order to obtain information. For instance, forklift truck drivers were one group that had a good overview of current production status and, therefore, represented an important source of information. The parquet scheduler even described the forklift truck drivers as "his tentacles into production". Having personal contact with the scheduler was also perceived as very important for first-line supervisors. During personal contact, the schedulers continuously and actively searched for or received information from other groups of employees. Through all sources of information, the scheduler actively paid attention to small pieces of information and reflected on consequences for the schedule. The schedulers were also expected to serve as information suppliers keeping up continuous communication with others in the organization. This formed an important basis for collaboration.

Many dialogues were made in a "humorous spirit" and one scheduler explicitly pointed out that he tried to cheer up others. By having a good relation with other groups of employees, he stated that he could more easily gain their support in case of troubles with certain production orders. The schedulers were expected to adapt to specific situations within or outside the company and recognize and consider requirements to manage problems and consequences of their scheduling decisions on other departments or group of employees. For example, the scheduler should generate schedules taking into account the consequences for other employees' working hours.

5.2. Technological aspects

Technology in the scheduled production system, described here as the primary technical system, limited the effectiveness of scheduling. The schedulers had to deal with uncertainties or technical design failures in the production process that restricted scheduling choices. For example, the schedulers in both the sawmill and the parquet manufacturer must deal with uncertainty regarding the outcome of production with respect to quality and volume until final production results were known. This was due to quality-grading procedures at the last stages in the production processes. Schedulers were obliged to make assumptions and approximations, mainly based on experience, instead of working with exact figures.

There were also examples of technical design constraints, for example at the sawmill where a large old drying kiln was run in parallel with two new drying kilns. Those produced superior results but had to be shut down for capacity regulation since the old kiln only worked at full capacity and the new kilns could not entirely replace the old one. The drying capacity was thus reduced compared to the "theoretical capacity", which strongly influenced the scheduler's decisions. Another example of the technology influencing the scheduling process was the technical failures in the automated production system at the furniture manufacturer, as the

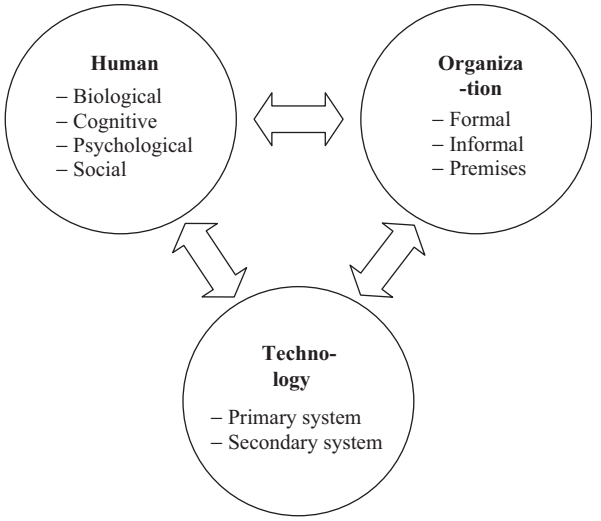


Fig. 1. The H-, T-, and O-components in the HTO-concept.

automated part reached only 50% of the planned capacity.

Another influence from technology was through the design of the scheduling software system, described here as the secondary technical system. All companies had standard scheduling software tools, but in all cases specific tools and spreadsheets had been developed to complement the standard tools and were used frequently. Spreadsheets were continuously developed by the schedulers to support scheduling. One important reason for developing the spreadsheets was the schedulers' great need for reliable data about the incoming material, production status and planned orders, data that could not be obtained from the scheduling software tool. There were examples in all companies studied of these spreadsheets being distributed to other groups of employees.

At the sawmill, a second reason for using the spreadsheets was the incompatibility between different computer systems. At the furniture manufacturer, a reason was that customer data were delivered on paper and could not be directly fed into the scheduling software system, and a further reason was that the scheduling software tool had not been adopted by all departments.

There were several reasons why the schedulers could not fully trust the data provided on the scheduling software system. Some well-known reasons for data inaccuracy such as mistakes in data entry were identified. Further to this, other reasons were identified. At the parquet manufacturer, the scheduler depended on reliable information from the sales department on preliminary orders; these were placed directly into the scheduling software system by the market representatives. However, the preliminary orders consisted of preliminary reservations that might not in the end be executed as customer orders. As the scheduler in this case needed more exact figures he tried to influence the market representatives to make final reservations, which was made by the scheduler delimiting the available stock in the scheduling software system.

In dealing with the scheduling software system, there were many examples of situations that required human interaction. At the sawmill, two incompatible software systems had to be used to optimize the high value output from the sawing process. The integration of the software systems' results was made by the scheduler. At the parquet manufacturer, the scheduling software system could not convert between parquet lengths and packages, which the scheduler was forced to do manually. At the furniture manufacturer, orders were received either manually or electronically and the scheduler had to coordinate these types.

5.3. Organizational aspects

In all the companies the schedulers participated in several regular meetings every week with customer representatives and production management. For example, the scheduler at the furniture manufacturer had weekly production meetings with the top management. The scheduler at the house manufacturer similarly had weekly production scheduling meetings with production management as well as delivery meetings with salesmen and project engineers.

In the studies, the schedulers worked in situations where the sales departments had high expectations of flexibility, short delivery times and responsive customer service and where, conversely, the production departments strove for more stable conditions. These departments both contributed to the overall aims of the company but followed entirely different logics concerning how these goals should be reached and how the different departments should be managed. The scheduling process took place in the middle of this goal conflict.

In all companies, the schedulers needed to coordinate and negotiate with these areas to realize schedules. A common argument in negotiations

was: "The other department agreed to this, will you help me with the rest?" indicating that the schedulers considered that both departments should take responsibility for solving problems. The need for this negotiation was not explicitly stated in the schedulers' task description, but the observations showed that in fact the schedulers provided a strong link between the production and sales departments. This aspect of the scheduling process, therefore, constituted an important part of the informal side of the organization.

In all companies, the schedulers had numerous contacts every day related to inquiries about for example feasibility in production for potential orders and information about changes. At the parquet manufacturer, the customer helpdesk secretary even moved into the scheduler's office in order to improve collaboration with the customer service department. By serving as an integrating link between production and the sales department, the schedulers were able to pass on information before it entered into the scheduling software systems or transfer information that did not exist at all in the scheduling software systems.

In all companies the schedulers were physically located in central locations, for instance close to production but at the same time well placed to be easily reached by employees from other departments and in two cases also by external suppliers. The location in conjunction with participating in meetings with management and representatives from various departments further facilitated the schedulers' role as an integrating link between the departments. However, at the house manufacturer, there were examples of the impact of a lack of proximity and collaboration between the scheduler and the sales department where extremely late changes in ordered products complicated the scheduling process.

6. Scheduling in an HTO-perspective

6.1. Human aspects influencing the scheduling process

The study showed that scheduling was characterized by high demands for flexibility, a need to deal with a high degree of uncertainty, and consisted of problem solving and multiple parallel tasks. These findings support those made in earlier research from other industries (McKay et al., 1992; Crawford, 2000, 2001; Jackson et al., 2004). These align with the findings from this study, which provides strong illustrations of the human contribution to the scheduling process at a cognitive level.

To fulfil the role as problem solver, the schedulers needed thorough detailed knowledge of the functioning of the production system. All schedulers had prior personal experience from working in production. Due to this personal experience, the schedulers gained legitimacy from "bottom-up" experience and had many strong links with people in the production departments. Their experience formed a personal knowledge base from which to carry out scheduling tasks and is, therefore, categorized at the psychological level of the human aspects. Also, the schedulers' legitimacy with others was based on the schedulers' ability to understand the production process, its limits and under what conditions work was executed.

Looking at human contribution at a social level, this study and earlier studies (Crawford, 2001), demonstrated the importance of the schedulers' frequent personal contacts with different groups of personnel and departments in the organization. One group leader at the parquet manufacturer even emphasized that "scheduling was not paper work carried out in the office." The schedulers thus acted as informal coordinators, that is as nodes through which information was spread and contacts between people were co-ordinated in the organization. The need to negotiate between incompatible goals, the expectations on schedulers to show social concern about consequences of the schedule for the employees, as well as the schedulers' dependency on others' willingness to help out in acute situations, were all examples of the strong demands placed on the schedulers' social abilities.

We may, therefore, conclude that the scheduling process may be influenced by the scheduler adding human capabilities that cannot be automated, such as extensive problem solving, and negotiating between and linking together groups of employees to handle incompatible goals. It follows that these individual qualities will also affect the outcome and quality of the scheduling process.

6.2. Technological influencing the scheduling process

This section reviews the interaction between the H- and T-components. Here the two main parts of the technical system are scrutinized separately.

The interaction between the production system and the scheduler can be described from different

angles. Previous research studies have identified that schedulers are involved in multiple interactions (McKay et al., 1995; Crawford, 2000; Karltun and Berglund, 2002b; Jackson et al., 2004) and that many devote a large proportion of their working hours to interacting with the production system (Wiers, 1997). In doing this the schedulers put a lot of effort into gathering information about the past, present and the anticipated future state of the system.

A complicating aspect to control is the high degree of uncertainty. Uncertainty was seen to emerge both from inside and outside of the company. Internal uncertainty can be construed as existing when something does not occur as the scheduler has assumed in his/her schedule. It is important to point out that the capability of the production system was determined at a detailed subsystem level, thus there was a theoretical maximum capacity. The ability to achieve to schedule in the companies studied was determined by the production units' abilities to reach this capacity and the schedulers' abilities to produce a feasible schedule, attempting to optimize the capacities in the different subsystems. The ability to achieve this may be constrained by circumstances, such as availability of raw material and practical/technical restrictions in parts of the machinery. These are examples of technology influencing individual behaviour and the scheduler's productivity (Pasmore, 1988).

The interaction between the scheduling software system and the scheduler is twofold. The system is a provider of information on the state of the production system. In some cases it is the sole provider. In this study, however, this information was only partially provided by the scheduling software systems. Necessary information had to be searched out by other means. As described before, this was in many cases achieved through visual inspection of production and stock and informal conversations. Walking around the facility and the conscious building of social networks within and outside the company were examples of measures used to support these activities.

The scheduling software system also acts as a decision support system. However, the available systems were criticized for providing insufficient decision support for the scheduling task. This deficiency has also been identified in earlier research on scheduling, in which data availability and accuracy, dealing with complexity, performance measurement, organizational embedding, and interaction with the human scheduler were inadequate (Wiers, 1997; Jackson et al., 2004).

The schedulers in this study had high expectations placed on them to provide information to others in the organization, information that in some cases was more detailed than that obtained from the scheduling software system. The inability of the systems to provide the type of information that the schedulers needed had resulted in development of spreadsheets that might be viewed either as workarounds or as powerful flexible aids to scheduling that complement the scheduling software system. Furthermore, in dealing with the scheduling software system, there were, as described in the findings, many examples of situations that required human interaction and support to sustain a scheduling process capable of achieving its set goals. The scheduling software systems in the companies studied could not fully support the schedulers in their tasks and did not comply with the design criteria regarding the HT-interaction in terms of information presentation, integration with the task, and providing relevant feedback.

In the companies studied, aspects of technology thus set prerequisites for the scheduling process in two key areas: the production system and its limitations, and the scheduling software system and its ability to provide sufficient data and control.

6.3. *Organizational aspects influencing the scheduling process*

Also it has been identified that the O-component created prerequisites for the scheduling process. The schedulers participated in several regular meetings with the management in the companies. In this way, the schedulers had a firm basis to contribute to key decisions having significant presence in the companies' formal meeting structures.

In the companies studied, there were several examples where the schedulers worked closely both with the production and sales departments to negotiate between the departments and handle their often-conflicting logics. There were strong links and interdependencies between the schedulers and many other employees often from unexpected groups such as the quality department and forklift truck drivers.

Schedulers possessed high-value information gained in their roles as integrating links and information nodes between various departments (Davenport et al., 1998; Jackson et al., 2004). As

others were dependent on this information, as well as the results of the schedules developed, the schedulers exerted considerable influence on the activities of the production, sales and purchase departments (including those of management). The schedulers thus exerted control without having the formal authority to do so, which is an important aspect of the informal side of the organization. This emphasizes the need to study real world scheduling practice, as only in this way can the true significance of the informal structure of the scheduling process be realized.

The study also showed examples of the importance of the physical location of the schedulers. Their physical proximity to others enabled them to more easily obtain information from production and other employees, and, therefore, to efficiently solve scheduling problems. This central physical location thus strongly contributed to their role as an information node and problem solver. In summary, organizational aspects influencing the scheduling process include: the degree of proximity between employees, meeting structures, the schedulers' position in hierarchy and their work role purpose in interconnecting activities of different organizational parts.

7. Conclusion

Comparing the H-, T- and O-components suggests that these are strongly interlinked aspects of scheduling. As this project had a focus on the scheduling work performed in practice and the schedulers' work role, the initial emphasis was on the H-component. The analysis revealed, however, the importance of analysing and understanding the T- and O-component as well. The case evidence and analysis presented in this paper supports many of the previous studies done on scheduling in practice. In particular, it supports many aspects of the model of human scheduling presented in Jackson et al. (2004).

All three components introduce conditions and limitations for the scheduling process, which strongly suggests that all three components need to be addressed when aiming at improving scheduling processes. These three components, when jointly investigated, established the key issues that had impact on the scheduling work studied. It became clear that the scheduler is part of an HTO-system, both being affected by it and affecting it. Particular issues of note concerning the *human* contribution are:

- the importance of the scheduler's personal experience of the production systems,
- the need for the scheduler to understand marketing, sales and company strategies as well as production and in daily work handle incompatible goals,
- the role of the scheduler as high-value information node,
- the need for the scheduler to have good social skills, and
- the role of the scheduler as a problem-solver.

This view complements the more traditional one that good schedulers should be logical analysts, by adding that there are many other requirements placed on them due to context-specific demands emerging from the real world environments.

Technology impacts the scheduling process as it sets prerequisites in two key areas:

- the production system characteristics and its limitations at a detailed level in terms of capabilities, design failures and reliability, and
- the scheduling software system and its ability to provide sufficient data, control and decision support as well as to support other aspects of the scheduling task that in this study were largely ignored.

The *organization* finally influences the scheduling process through both formal and informal dimensions:

- the formal organization of the scheduling process, including formal meeting structures,
- the scheduler's informal authority,
- the scheduler's role to interconnect activities between different organizational groups, and
- the degree of proximity between employees.

Further aspects of scheduling that may benefit from an HTO-analysis include distributed decision-making in the scheduling process as well as the scheduler's gaining control and coping with uncertainty.

We suggest that by taking the HTO components into account from the beginning of any study it is possible to better establish and visualize a broad

range of issues that contribute to the outcome of the scheduling process.

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